

MAT 524: Functions of a Real Variable II

Homework IV

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[[Problem One]]:

Problem Statement

Suppose $h \in L^p(E)$ and that f is a measurable function on E . If $\|h - f\|_p < \infty$, show that $f \in L^p(E)$.

Solution

Since $h, h - f \in L^p(E)$ and $L^p(E)$ is a vector space,

$$f = h + (-1)(h - f) \tag{1}$$

is in $L^p(E)$.

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[[Problem Two]]:

Problem Statement

Show that the step functions are not dense in $L^\infty([0, 1])$.

Solution

Let $A = \bigcup_{n=1}^{\infty} [\frac{1}{4n+1}, \frac{1}{4n-1}] \subset [0, 1]$. The characteristic function χ_A oscillates between 0 and 1 infinitely many times, due to the nature of the set A . Note that there is no step function that can mimic this behavior, since, by definition, step functions can only change values finitely many times on any interval. With this in mind, let s be any step function on $[0, 1]$. On the interval $[0, 1/3]$, there must be some interval of positive measure where the distance between s and χ_A is greater than or equal to 0.5, i.e., $\|\chi_A - s\|_\infty \geq 0.5$. Thus, any ε challenge with $\varepsilon \leq 0.5$ will fail, meaning the step functions are not dense in $L^\infty([0, 1])$.

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